What a low-cost smartphone protocol can and cannot establish about urban noise

Hanoi, 363 field measurements, three negative results
and the instrument we ended up building

Laurian Jamin **Lucas Zborowski**

Research interns, ISIMA — Clermont-Ferrand, France

COSMOS Lab, VinUniversity — Hanoi, Vietnam

Supervised by Doanh Nguyen-Ngoc · With Nguyen Thanh Quang

June – August 2026

Context and research questions

The gap this project sits in

- Hanoi: dense, fast-growing, **motorcycle-dominated** traffic
- **No national strategic noise-mapping programme** and no statutory noise action plan in Vietnam
  Nguyen et al. 2025, *City Environ. Interactions*
- **One** instrumented campaign ever published for Hanoi — and it measured in **2005–2007**
  Phan et al. 2010, *Applied Acoustics* 71(5)
- Class 1 sound level meters and commercial propagation software: out of reach for most local research budgets

The question

How far does a protocol built from **three consumer smartphones** and **open data** actually get you?

The honest answer turned out to be: further than nothing, and much less far than a map.

The research question, and what is out of scope

The question, as we posed it in June — unchanged

Can urban noise be attributed to **the form of the city**, and to **the type of vehicle** passing through it? Models validated on car-centric European cities are unproven on motorcycle-heavy soundscapes.

How it was operationalised

1. Can OpenStreetMap morphology predict noise at an **unmeasured location**?
2. Does a model **transfer** between cities?
3. Does vehicle **flow from video** explain the levels?
4. What can the protocol claim with **no absolute reference**?

Out of scope, declared up front — night-time (10 points after 21:00), compliance against any standard, any prediction beyond the three sites.

And the answer, on both halves

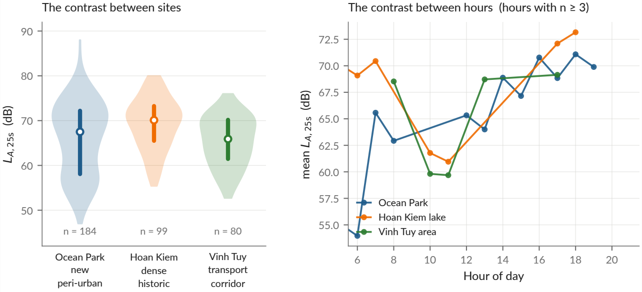
Form: no — morphology adds nothing the strict protocols can see.

Vehicle type: no — and the chain that was meant to measure it does not work.

What survives is $\log(d_{\text{road}})$: distance to the road, and little else.

Data and protocol

The campaign



`data/processed/measurements.csv`

Protocol — 3 handsets, *Decibel X* (A-weighted, SLOW), 20–30 s per point, a clip $\geq$ 10 s, a 10 m GPS gate, ODK Collect $\rightarrow$ Kobo.

Published, CC-BY-4.0 — both CSVs and both XLSForms. Videos and raw Kobo exports are **not**: faces, plates, collector identities.

Metrology — the constraint everything else is built around

Three phones, cross-calibrated against *each other* and against nothing

A bias common to all three is **invisible in the data by construction**.

What that forbids

- Stating an absolute level as a fact
- Any compliance claim under QCVN 26:2010 — it requires a class 1 or 2 instrument

What it still allows

- Contrasts *between places* and *between hours* — a common bias cancels in a difference
- A *bounded* bias interval from instrumented literature: $+3.0$ **to** $+7.3\,\mathrm{dB}$

The measured quantity, and the only one:

$$L_{A,25\mathrm{s}} \;=\; 10\log_{10}\left(\frac{1}{T}\int_0^T \frac{p_A^2(t)}{p_0^2}\,\mathrm{d}t\right), \qquad T \approx 25\,\mathrm{s},\ p_0 = 20\,\mu\mathrm{Pa}$$

Not $L_{\mathrm{Aeq,\,8h}}$, **not** L_{den}, **not** L_{night} — those integrate over hours or a year.

`docs/metrology.md` · `results/tables/literature_anchoring.md` the interval is estimated, never applied

Method and evaluation protocol

The evaluation protocol — and why it is the whole story

Every model is scored under **three splits**, on exactly the same folds:

1. **Block-CV, 600 m** — square blocks in UTM 48N, `GroupKFold` on the block. The permissive one.
2. **Buffered leave-one-out, 300 m** — for each point, train on *all* points more than 300 m away.
   The reference protocol.
3. **Leave-one-site-out** — generalisation to an unseen urban typology.

Every score carries a 95 % interval bootstrapped **by block**, not by point: resampling correlated points gives an interval that is far too narrow.

Why 300 m and not less

The morphology features aggregate over a disc of **radius 300 m**. Two points 110 m apart still share **77 %** of that disc — and two either side of a cell boundary share **almost all** of it.

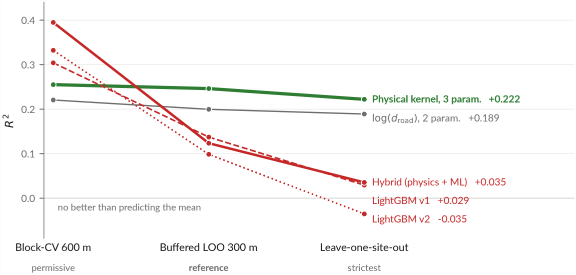
Group on 110 m cells and the model sees near-twins of its test points in training. The score is then not out-of-sample.

Roberts et al. 2017, *Ecography*

`scripts/04_evaluate_models.py`

Results

The ranking inverts — and that is the result



`models/metrics.json -- the same folds for every model`

Read across, not down

The elaborate architecture **wins the permissive split and loses both strict ones.** Three parameters is the only flat line.

What we did about it

The hybrid is **tested and rejected at this sample size** — not deferred to future work.

The delivered model

A **line**-source attenuation kernel — energy falling as $1/d$, not $1/d^2$: a street is a line of sources, not a point.

$$E = \frac{A_{\mathrm{hw}}}{\max(d_{\mathrm{hw}}, D_0)} + \frac{A_{\mathrm{res}}}{\max(d_{\mathrm{res}}, D_0)} + B$$

$$L = 10 \log_{10} E$$

$A_{\mathrm{hw}} = 4.774 \times 10^7$ $A_{\mathrm{res}} = 3.800 \times 10^7$ $D_0 = 5\,\mathrm{m}$

- $-3\,\mathrm{dB}$ per doubling of distance, not -6
- Positivity-constrained: $A_{\mathrm{hw}}, A_{\mathrm{res}}, B > 0$
- ***B* fits to** $-99.6\,\mathrm{dB}$, a boundary solution. **The background term is not identifiable here.**

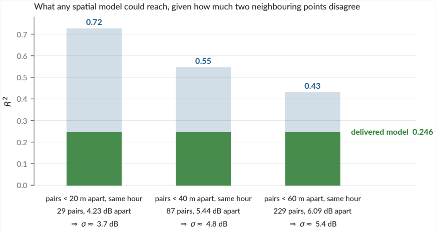
`models/hybrid_physical.json` · `models/metrics.json`

Performance under the reference protocol

R^2	**0.246**
95 % CI	[0.097, 0.339]
RMSE	6.20 dB
MAE	5.01 dB
data spread (σ)	7.14 dB

Say the error out loud

A typical prediction is wrong by **5 dB** — a factor of three in energy. Enough for a contrast, nowhere near an assessment.



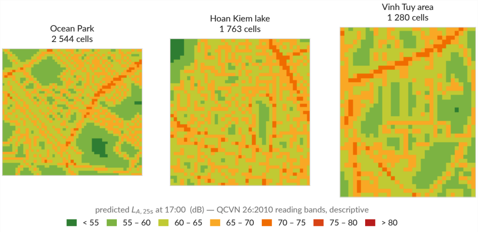
`data/processed/measurements.csv` · `docs/metrology.md`

Two points a few metres apart *at the same hour* must be predicted identically. Whatever they disagree by is variance **no spatial model can explain**: $\Delta L \sim \mathcal{N}(0, 2\sigma^2) \Rightarrow \sigma = \frac{\sqrt{\pi}}{2}\mathbb{E}|\Delta L|$, and $R^2_{\text{max}} = 1 - \sigma^2/s^2$.

At the 40 m scale the map works at

The ceiling is near $R^2 \approx 0.54$. The delivered **0.246** is about **half** of what is attainable, not a tenth.

The map — and the caveat that goes with it



`results/maps/hanoi_noise_map.csv -- 5 587 cells`

The grid step is not the accuracy

40 m cells from a model whose typical error is **5 dB**. Bands are QCVN 26:2010, **descriptive**.

And no hour, either

17 hourly columns, **all 17 identical**: the kernel has no hour term. **Nothing varies that the model does not model.**

The three negative results

1 — A three-parameter physical law beats every learned model we built

Six-variable gradient boosting, engineered features, a hybrid architecture: all lose the reference protocol to two distances and a constant. At $n = 363$ over three sites, there is not enough signal to justify capacity.

2 — Cross-city transfer fails, and in three distinct ways

Pretrained on Sunbird Uganda, scored on our 363 points: $R^2 = -15.9$ (v1), -8.2 (convention-invariant v2). **Not a calibration offset** — removing the 26 dB mean difference still leaves R^2 negative. **v1 transfers anti-information**, $r = -0.38$: it ranks quiet and loud the wrong way round. **v2 leaves nothing**, $r = +0.08$ — not wrong, empty. Against which $\log(d_{\mathrm{road}})$ alone reaches **0.240** on the same points.

3 — Vehicle flow does not explain the levels

Non-negative energy regression on 147 matched videos returns **zero coefficients** for motorcycles and cars. Speed and source–receiver distance are not observable from a non-georeferenced camera. **Section 4 revisits how much of this is physics and how much is instrument.**

`make transfer` · `scripts/experiments/uganda_transfer.py` · `docs/negative-results.md`

Auditing our own data

We found one of our own published results was wrong

Retracted, August 2026

A LightGBM model scored $R^2 = 0.45$ under what was then described as honest spatial cross-validation. A GAMA simulation was built on it. A project-update deck presented the figure.

The grouping was on **110 m cells** while the features aggregate over **300 m**. It leaked.

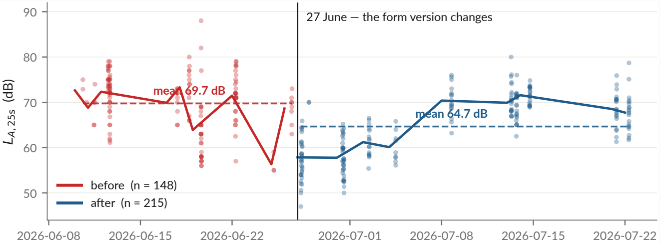
What replaced it

$R^2 = 0.246$ under buffered leave-one-out — roughly half the withdrawn figure, and the one that survives a split excluding the feature support.

- The retracted deck and map are in `docs/archive/`, **with their reason**, not deleted
- Nine corrections applied the same day
- The project pivoted from *producing a noise map* to *studying what this protocol supports*

`docs/audit/scientific-audit.md` · `docs/project-timeline.md` · `docs/archive/`

Finding 1 — a $-5.1\,\mathrm{dB}$ discontinuity on 27 June



`data/processed/measurements.csv` · `docs/audit/`

Controlling for site, hour and class: $-5.14\,\mathrm{dB}$ (SE 0.55, $p \approx 10^{-20}$). The form version changes the same day — integer readings 87 % → 20 %.

Why it matters

Before/after is predictable from $(\mathrm{lat}, \mathrm{lon})$ alone at **AUC 0.83**. An instrumental shift is then spatially structured, and a model can learn it **as morphology**. **Status:** **open**.

Findings 2 and 3 — the video chain, and the intervals

2 — The video chain is measuring something, but not what we think

- Detector modal split: **57 % cars, 36 % motorcycles**. In Hanoi. Reality is roughly the inverse.
- Video→measurement gaps: median 15 s, p90 56 s, **max 161 s** — for a 25 s sample
- Matched subset spans 61–80 dB against 47–88 dB overall: σ **cut by 42 %**
- Flow vs level: $r = -0.11$. A *negative* correlation between traffic and noise is not physical.

Consequence for negative result 3: it is far more likely an **instrument failure** than a finding about acoustics. It should be reported as *"the chain could not be made to work"*.

3 — The headline ranking is not statistically separated

`physical` $R^2 = 0.246$, CI $[0.097, 0.339]$ vs `dist_road` $R^2 = 0.200$, CI $[0.077, 0.266]$. The intervals overlap almost entirely. **"The physical kernel beats log-distance" is a coin flip, and the deck's own table should show the CIs.**

What the audit leaves standing

Holds

- The evaluation protocol and the ranking inversion
- Negative result 1 — capacity does not pay at $n = 363$
- Negative result 2 — transfer fails, $R^2 < 0$
- Provenance: raw $\rightarrow$ published is lossless and checkable
- Every published number traces to one file

Needs work before submission

- The 27 June discontinuity — diagnose, then refit with the phase
- Negative result 3 — reframe as an instrument failure
- Carry the confidence intervals into every summary table
- Flag observed vs derived values in the published CSV (15 % of `construction_nearby` is inferred from the noise class)

The instrument was the problem in July, and it is still the problem in August.

Which is the argument for the next section.

From a pipeline to an instrument

Why build an application at all

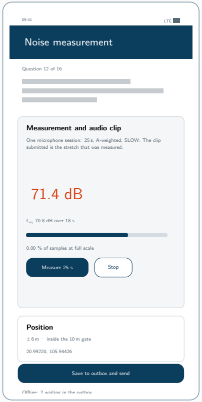
Four things the campaign could not do

1. **Control the instrument.** *Decibel X* is closed: no visible weighting, window or gain.
2. **Measure and record in one session.** Two apps meant the level of one stretch of street and the audio of another.
3. **Scale beyond three people.** Nobody pairs ODK Collect and a sound meter by hand.
4. **Record what the device was.** Handset, OS, audio source — none of it reached the dataset.

The decision, 19 August 2026

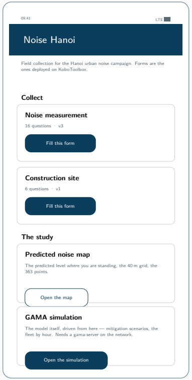
One application that **replaces the pairing** — forms, level, clip, GPS fix, submission, offline-first — and **carries the study**, negative results included.

`mobile/README.md` Kotlin + Compose, one `:app` module, `minSdk` 26



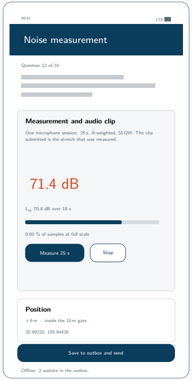
one session: level + clip

The application



Collect

both forms, offline

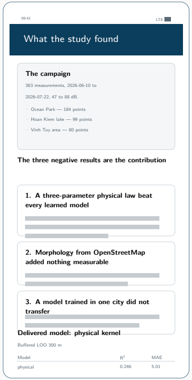


Measure

25 s, A-weighted

Map

the level where you stand



Results

read from `metrics.json`

Vector mock-ups drawn from the Compose source. **No device capture exists yet** — which is the last slide of this section.

What the app measures — and what it refuses to claim

The app's level is a *different quantity* from the campaign's

This handset's microphone, not a trimmed Decibel X reading. It goes in **its own field**, with the device and the audio source the platform granted. **The two never share a column.**

Verified end to end, live

- Fix, 25 s measurement, instance written, **submission accepted by** `kc.kobotoolbox.org`
- A position outside the three areas is **refused, not answered**
- 36 unit tests; weighting pinned to the IEC table

Not verified — and it is the important one

The level has never been checked against a sound source. An emulator records silence. Until it is compared to a meter, **the absolute figure means nothing** — equally true, per `docs/metrology.md`, of the campaign's own numbers.

First task on a real handset, alongside Decibel X on the three campaign phones.

Driving GAMA from the phone — how it is wired

One socket, and the phone runs nothing

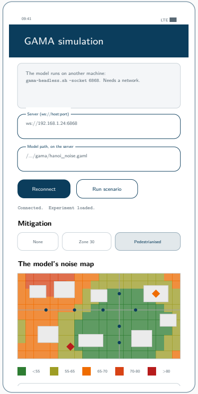
No Android port — but a **WebSocket API**: `gama-headless.sh -socket 6868`. The app is a client (`load`, `play`, `step`, `ask`, `stop`): a remote control and a screen.

Two things only driving it teaches you

A `type: gui` experiment **kills the socket** on load, and `step` without `sync` advances nothing while **every indicator reads zero**.

It redraws the model's display, it does not invent one

Every layer and colour comes from the aspect blocks of `hanoi_noise.gaml`, **in declaration order — which in GAMA is the stacking**.



ARCHITECTURE.md §5.2

Next: Clermont-Ferrand

Still no reference instrument

There will be no sound level meter at Clermont either. The calibration gap does not close, and nothing said here should suggest it will: the app's level stays relative, and the half-day next to a class 2 meter remains *unbooked*, not *done*.

So what *is* the work

Maintenance and deployment, not new instrumentation. Keep the application alive, then get it collecting in a second city.

`docs/handover.md` §6, debt #2, P0-3

What the second city is actually for

- **One instrument, two cities.** The **same APK** in Hanoi and Clermont. That is the point: the instrument stops being a confound, which is the one thing the Uganda transfer could never fix.
- **Two fabrics as different as we could pick.** Two-wheelers against cars and a tram; dense organic against European planned; flat delta against volcanic relief.
- **A fourth typology, at last.** Three sites bounded everything here. A second city is the only thing that moves that number.

The comparative deployment — and why it is the decisive test

	Hanoi	Clermont-Ferrand
Fleet	two-wheelers	cars, tram
Fabric	dense, organic	European, planned
Relief	flat delta	volcanic, hilly
Monitoring	none	EU END mapping

Held constant — the **same APK**, form, accuracy gate, $L_{A,25\mathrm{s}}$, OpenStreetMap features and evaluation protocols.

Why this is stronger than Uganda → Hanoi

There, the **instrument, protocol and feature conventions all differed too**: "transfer fails" and "the datasets are not commensurable" are not separable. One app in both cities and **the instrument stops being a confound**.

And a ground truth, finally

Clermont-Ferrand falls under EU END strategic mapping — the first **official, instrumented reference map** to compare against.

The two things a reviewer will press on

1 — No absolute reference. Was half a day really out of reach?

Three handsets calibrated against *each other* and against nothing. `docs/metrology.md` owns this squarely — but an acoustics reviewer will not ask whether we *knew*; they will ask whether **one side-by-side reading against a class 2 meter** was genuinely unobtainable.

It is the right question, and the honest answer is that a **half-day rental** would settle it. **One reading turns "calibrated in relative terms" into "calibrated"**, and every level inherits it. The cheapest item anywhere in this work.

2 — Three sites are three typologies, and that binds harder than the R^2

Our own text says it: the number of **independent morphological configurations** is close to **three**, whatever the number of points. $n = 363$ does not buy $n = 363$ worth of generalisation.

The hard limit is the design, not the R^2. **A fourth and fifth contrasting fabric would do more than any modelling change.**

Conclusion

What this work established

Findings

1. At $n = 363$ over three sites, **three physical parameters beat every learned model** we built. Capacity does not pay.
2. **Cross-city transfer fails** — $R^2 < 0$, with matched features.
3. The video chain **could not be made to work**; that is an instrument result, not an acoustic one.
4. A cross-calibrated smartphone protocol supports **contrasts**, never levels.

Practices that made those findings trustworthy

- The buffer matches the feature radius — always
- Baselines and an ablation on *the same* splits
- The delivered model is chosen **by code**
- Every number has **exactly one home**
- Retracted work is archived **with its reason**
- Nothing is predicted outside the sampled envelope

The deliverable is a **protocol, an instrument and an honest error bar** — not a noise map of Hanoi.

Thank you

github.com/Colauz/noise-modelling-hanoi

Everything on these slides rebuilds from the repository:

`make features`   `make models`   `make results`   `make report`

the deck's own figures: `scripts/12_presentation_figures.py`

COSMOS RESEARCH LABORATORY WELCOME COHORT 5

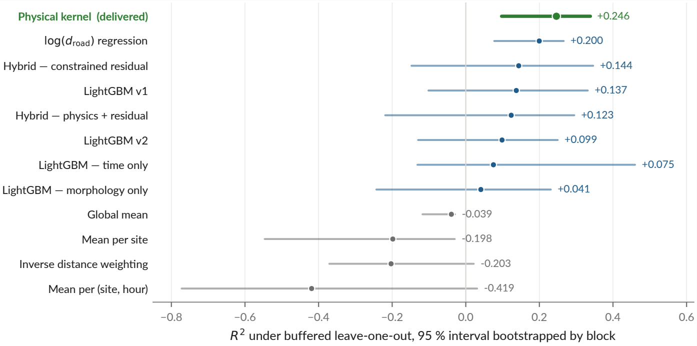
VINUNIVERSITY

CLERMONT AUVERGNE INP Isima

Backup — every formula this talk rests on

What	Definition
The measured quantity	$L_{A,25\mathrm{s}} = 10\log_{10}\left(\frac{1}{T}\int_0^T \frac{p_A^2(t)}{p_0^2}\,\mathrm{d}t\right), \quad T \approx 25\,\mathrm{s}, \;\; p_0 = 20\,\mu\mathrm{Pa}$
A-weighting, IEC 61672	$A(f) = 20\log_{10}\frac{12194^2 f^4}{(f^2+20.6^2)\sqrt{(f^2+107.7^2)(f^2+737.9^2)}\,(f^2+12194^2)} + 2.00$
Levels never average	$L_\Sigma = 10\log_{10}\sum_i 10^{L_i/10}$ decibels add in energy — which is why everything below works on E
The delivered model	$E = \frac{A_{\mathrm{hw}}}{\max(d_{\mathrm{hw}}, D_0)} + \frac{A_{\mathrm{res}}}{\max(d_{\mathrm{res}}, D_0)} + B, \quad L = 10\log_{10} E, \quad A_{\mathrm{hw}}, A_{\mathrm{res}}, B > 0$
Why $1/d$, not $1/d^2$	A street is a **line** of sources, not a point: $-3\,\mathrm{dB}$ per doubling of distance, not -6
The traffic scenario	$L(k) = 10\log_{10}\left(k\,(E - E_{\mathrm{res}}) + E_{\mathrm{res}}\right)$ **not** $L + 10\log_{10} k$
Reference protocol	For each point i, train on $\{j : d(i,j) > r\}$ with $r = 300\,\mathrm{m}$ = **the feature aggregation radius**
The ceiling on R^2	$\Delta L \sim \mathcal{N}(0, 2\sigma^2) \Rightarrow \sigma = \frac{\sqrt{\pi}}{2}\,\mathbb{E}\lvert\Delta L\rvert, \quad R^2_{\max} = 1 - \sigma^2/s^2$

Backup — every model, with the interval that qualifies it



`models/metrics.json -- 2 000 resamples, bootstrapped by block`

This is audit finding 3, drawn

`physical` and `dist_road` overlap almost entirely. "The physical kernel beats log-distance" is a **coin flip** at this sample size. Other protocols: `models/model_comparison.md`.

Backup — the absolute-bias interval

Anchor	Instrument	Quantity	Raw gap	Corrected
Gelb & Apparicio 2019, HCMC	dosimeters + GPS	$L_{\mathrm{Aeq,1min}}$	+9.3	+7.3
Phan et al. 2010, Hanoi	RION NL-21/22	L_{den}	+7.0	+3.0
Phan et al. 2010, Hanoi	RION NL-21/22	L_{night}	−3.0	−3.0
IOHE, 12 arteries (press)	undocumented	daytime mean	+8.6	+7.6

Conclusion

Plausible absolute bias: **$+3.0$ to $+7.3\,\mathrm{dB}$** (peer-reviewed sources, night excluded — $n = 10$ is too thin).
Corrected gap = raw gap minus the difference the quantity alone explains: it approximates the residual instrumental bias, and **is reported and never applied**.

`results/tables/literature_anchoring.md` · `scripts/06_anchor_literature.py`

Backup — what is published, and what is not

Dataset	Size	Published	Why / licence
Field measurements	363 pts, 3 sites	yes	CC-BY-4.0
Vehicle counts from video	147 videos	yes	CC-BY-4.0
Survey forms (XLSForm)	2 forms	yes	CC-BY-4.0
Traffic videos	147, 6.0 GB	no	faces and plates
Raw Kobo exports	—	no	collector identities
OpenStreetMap extract	49 146 buildings	no	regenerable; ODbL
Sunbird Uganda 61K	61 k clips	no	gated upstream

Stated plainly in the repository

No ethics review was requested for the video collection, and none is claimed. Only non-identifying aggregates are released. The retention decision is still **open**, and is a supervisor decision.

`docs/data-sources.md` · `LICENSE, LICENSE-DATA`